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How do teachers spend their time? A study on teachers' strategies of selection, optimisation, and compensation over their career cycle

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HIGHLIGHTS

- Workload-data from 1939 teachers from a representative school sample show that:
- Teachers invest their time resources differently according to their age.
- Younger and older teachers engage in fewer non-teaching tasks (selection).
- Older teachers invest free time resources in less demanding tasks (optimisation).
- Older teachers have fewer career ambitions (compensation).

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ABSTRACT

In order to address all teaching-related tasks, teachers often work long hours. Effective resource allocation is, therefore, invaluable if they want to manage their workload and remain healthy. This prompted us to assess how teachers allocate their time to different tasks over their careers. Results of a study involving 1939 German teachers from a representative school sample show that beginning teachers and those at the end of their careers engage in fewer tasks (selection). The latter also save time on demanding aspects and invest the time saved in less demanding tasks instead (optimisation), and have fewer career ambitions (compensation).

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1. Introduction

One of the concerns of school policy in many Western countries is how to retain effective teachers in schools (OECD, 2005). This is indeed an important task, bearing in mind that teaching is known to be associated with a high risk of burnout (e.g., Farber, 1991; Rudow, 1999). It has been shown that once teachers suffer from burnout, their job commitment (Hakanen, Bakker, & Schaufeli, 2006), as well as their overall well-being and health (Milfont, Denny, Ameratunga, Robinson, & Merry, 2008) may decrease considerably. Burnout is characterised as a syndrome consisting of three components: emotional exhaustion, depersonalisation and feelings of reduced personal accomplishment (Maslach, 1998). The present study focuses on the first component, emotional exhaustion, which is considered a key component of burnout (Maslach, 1998) and a central variable for understanding the burnout process (Cropanzano, Rupp, & Byrne, 2003). It describes a state of overexertion and depletion of one's emotional and physical resources (Maslach, 1998) and refers to the strain dimension of burnout.

A great number of self-report studies with teachers from a range of different countries have shown that stress and burnout of teachers are associated with high workload and time pressure. Results of the EUROTEACH study, a cross-sectional multi-centred study with 2796 teachers from 13 European countries (among them Britain, the Netherlands, France, and Italy), highlight the importance of time pressure as one of the major demands faced by teachers (see Verhoeven, Maes, Kraaij, & Joekes, 2003). Another study conducted by Hakanen et al. (2006) highlights the relevance of perceived workload to the burnout levels of 2038 Finnish teachers. Other studies from different countries have confirmed that perceived workload is demanding. Collie, Shapka, and Perry

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(2012), for example, report that workload-related stress decreases job satisfaction in 664 Canadian elementary and secondary school teachers. In his overview, Rudow (1999) sums up that workload is a major demand on teachers which contributes to their feelings of emotional exhaustion.

Definitions of workload need to take into account that in addition to objective demands (i.e., working hours), individual employees differ in terms of their subjective experience (i.e., perceived workload) and their coping skills (Hart & Staveland, 1988; Mejman & Mulder, 1998). A variety of factors, such as type and level of demands, personal discretion allowed (decision latitude), level of knowledge and (self-regulation) skills as well as current psychological state, will determine how well an employee copes with the situation (Mejman & Mulder, 1998). A study by Skinner and Pocock (2008) of 887 employees has, however, demonstrated that working hours and control over the work schedule as well as subjective workload are significant predictors of well-being. In this paper, we will thus focus on working hours as an objective work demand on teachers and secondly on the subjective aspects of how teachers manage their time effectively.

2. Long working hours as a demanding aspect of teacher work

There is some recent empirical evidence that the workload of teachers in terms of working hours is indeed high. In the UK, teachers without management responsibilities work 50 h or more per week during term time (Butt & Lance, 2005; Office of Manpower Economics [OME], 2005) on average. In comparison, UK managers and other professionals work 45 h during a regular working week (PriceWaterhouseCoopers, 2001). Equally high figures are reported for teachers from the US (Bruno, Ashby, & Manzo, 2012) or New Zealand (Ingvarson et al., 2005) and Lacroix, Dorsemagen, Krause, and Bäuerle (2005) show that the situation is similar for teachers from Germany (see Table 1).

Teachers' job profiles are very complex. Obviously, a considerable amount of teachers' working time is spent on teaching as the core task of teachers. According to the OECD (2011), the percentage of teaching time in relation to teachers' total statutory working time over all levels of education was 46% in 2009 (US: 54.7%, UK: 54%, Germany: 42.7%). This part of teachers' work is formally specified in most countries (OECD, 2011). However, class teaching time is only one aspect of teachers' highly complex job profile. In addition to teaching, they have a wide range of other tasks to fulfil (OECD, 2011). Some of these tasks, e.g., preparing engaging lessons and correcting tests and homework, are closely related to teaching and form a crucial aspect of being a teacher. However, apart from these tasks, teachers also need to fulfil a great number of additional tasks (e.g., administrative tasks, organising excursions or school projects). How teachers allocate their time to these tasks is less formally regulated in many countries (OECD, 2011). A differentiated study in the UK by Gunter et al. (2005) showed that teachers in secondary school spend 20% on supporting tasks (i.e., planning and preparing lessons, tests, homework, keeping records of pupil performance), 11% on other pupil contact, 6% on school/staff management and general administration respectively, and 13% on other duties in addition to the 44% of their working time dedicated to teaching. Bauer et al. (2007), who examined the weekly working hours of 949 teachers from Germany, arrive at a similar result. Whereas preparing lessons is the most time-consuming activity, followed by correcting coursework, communication with pupils and parents, administrative duties and other tasks, the least time consuming activities are project work and supervision of students. This complexity of teachers' job profiles is often not explicitly recognised, which may add to teachers' stress and burnout because of possible uncertainty as to who is responsible for what (OECD, 2005).

Furthermore, research indicates that there are particular tasks that may contribute to teacher stress and burnout. Tasks closely related to teaching (e.g., preparing lessons, correction of tests), as opposed to additional tasks (e.g., administrative tasks) may differ in their impact on teacher stress and burnout. Tasks more closely related to teaching might be relevant to teacher burnout because of the higher amount of time they require. Results from Smith and Bourke (1992) support this assumption. The 204 Australian teachers in their study report that assessment was, in their opinion, the most demanding aspect of their job whereas involvement with school-community as an additional task was perceived as only slightly demanding.

Other studies showed that teachers perceive additional tasks as especially demanding although they take up less time than tasks closely related to teaching. Dunn and Shriner (1999) presented 136 teachers with a list of 15 tasks and asked them to rate how enjoyable they found these tasks. The teachers reported that committee work (as aspect of their administrative duties) was the least enjoyable of all 15 tasks listed. Results by Chaplain (2008) also indicate that additional tasks and administrative tasks (e.g., paperwork) in particular are perceived as demanding. In his overview, Kyriacou (2001) also highlights that teachers perceive administrative tasks as particularly demanding aspect of their job.

To sum up, the studies reported in the introduction show that teaching is associated with high stress levels and a high risk of burning out (Farber, 1991; Rudow, 1999). We are going to investigate three aspects: teachers' working time (in line with Skinner & Pocock, 2008), the complexity of teachers' job profiles (addressing concerns of the OECD, 2005), and engagement in certain tasks (especially administrative duties; in line with Chaplain, 2008; Dunn & Shriner, 1999; Kyriacou, 2001), which may all be relevant to teacher burnout. We intend to confirm previous findings on the basis of a large sample of teachers from German secondary schools and test whether these three aspects are associated with emotional exhaustion as the core component of teacher burnout (Maslach, 1998).

This contributes to research insofar as many studies on workload apply self-report scales of demands and burnout and, thus, face the problem of a common-method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). We focused, however, on objective task demands and thus used instruments other than teacher self-reports on demands scales by providing a list of tasks (e.g., individual lesson planning, meetings with students or parents, administrative tasks such as paperwork, documentation) and

Table 1
Empirical studies on teacher working time from different countries.

Country	Sample	Hours worked in an average week	Source
UK	1052 class teachers (57% primary, 43% secondary school)	52	OME (2005)
	477 secondary school teachers	49.9	Butt & Lance (2005)
US	983 public schools teachers (60% elementary, 29% high school, 11% Kindergarten)	54 (58 incl. weekends)	Bruno, Ashby, & Manzo (2012)
Germany	20–9129 teachers, 14 different samples	49.4 (range 45.0–57.3)	Lacroix, Dorsemagen, Krause, & Bäuerle (2005)
New Zealand	1150 teachers from, wide range of schools	47	Ingvarson et al. (2005)

asking teachers to rate how many hours they spent on each task. Using this approach, we expect the correlations between the amount of working time and emotional exhaustion to be lower than in other self-report studies. These correlations will, then, give us a less biased indication of how demanding working hours, job complexity and particular tasks really are. Based on these results, we examine how much of their time teachers spend on the tasks associated with teaching and whether teachers in different phases of their careers allocate their time differently.

3. Self-regulation as a personal resource of teachers

What is more, after confirming this relationship, it becomes crucial to investigate how that part of teachers' working time when teaching-related tasks must be addressed should be organised. Ingvarson et al. (2005) summarise three areas for improvement. Firstly, at the system level changes need to be made in curricula or systems of recognition/reward for effective teachers. Secondly, at the school level, a professional culture should be established and student behaviour management needs to be implemented. Thirdly, at the individual teacher level, variation in individual teachers' capacity to manage work demands effectively needs to be addressed. This argumentation is in line with the job demands-resources model (JD-R; e.g., Schaufeli & Bakker, 2004), a current psychological model on occupational health and well-being. The JD-R-Model distinguishes between different demands and resources on either the job/task level (Ingvarsons points 1 and 2) or the individual teacher level (point 3).

This paper focuses on the third aspect – teachers' individual resources – and investigates teachers' ability to self-regulate as an essential personal resource (Lord, Diefendorff, Schmidt, & Hall, 2010). In particular, we address the research question of how teachers can manage their working time effectively (as a subjective aspect of workload). To do so, we draw on the conservation of resources (COR) theory by Hobfoll (2001, 2002) and the selection, optimisation, and compensation (SOC) model (Baltes & Baltes, 1990; Lang, Rohr, & Williger, 2011).

3.1. Conservation of resources (COR) theory

COR theory represents an integrated resource model which describes resources as "part of a greater dynamic process associated with well-being through the general use of resources" (Hobfoll, 2002, p. 311). COR theory states that individuals strive to obtain, retain, and protect their resources (i.e., free time, stamina/endurance, feeling that one is accomplishing one's goals and many more; for an overview, see Hobfoll, 2001).

According to Hobfoll (2002), stress and burnout occur when individuals lose these resources, are threatened with their loss or, fail to gain new resources after substantial resource investment. COR theory has also been applied to the work context. Hobfoll (2001) describes burnout as a result of the latter, i.e., a lack of resource gain following significant resource investment of time and energy. Teachers with their long working hours, for example, have already invested a substantial amount of time and resources in their jobs and may not be able to gain new resources which would help them to manage the demands of their work on the long run. As a consequence, there is a risk of eventual burnout. Investing their resources differently and maybe more effectively, however, may protect them against (further) resource loss, help them to recover from loss and gain new resources (a so called gain cycle might develop; Hobfoll, 2002).

It is thus crucial to investigate strategies by which teachers can allocate their time effectively (e.g., spending less time on demanding tasks and allocating free resources to less demanding aspects) in order to avoid emotional exhaustion and to remain healthy. As a framework for this, we draw on selection, optimisation, and compensation (SOC) theory (Baltes & Baltes, 1990; Lang et al., 2011).

3.2. Selection, optimisation, and compensation (SOC) theory

The coordinated use of selection, optimisation, and compensation can not only increase one's resources but it also helps individuals to maintain functioning when faced with demands, and to regulate impending losses in resources (Baltes & Heydens-Gahir, 2003). SOC can thus be characterised as an essential self-regulation strategy of individuals (Lang et al., 2011). Freund and Baltes (2002) argue that SOC strategies facilitate optimal resource allocation and thus contribute to successful adaption to (work) environments and in turn to well-being.

Firstly, *selection* involves setting or prioritising goals or tasks for which resources are available or can be obtained (Lang et al., 2011). These tasks should be in accordance with personal needs or environmental demands (Wiese, Freund, & Baltes, 2000). Selection can either be guided by personal preferences (so-called elective selection; Freund & Baltes, 2002), e.g., when individuals choose to focus on those aspects of their (working) life which they consider most interesting or, faced with a loss of resources which threatens their level of functioning, need to focus on primary goals and tasks (so called loss-based selection; Freund & Baltes, 2002). Selection processes help individuals to channel their development (Baltes & Heydens-Gahir, 2003).

Secondly, *optimisation* refers to an individual's ability to acquire and refine the means to achieve selected tasks or goals; such means could be practice, acquisition of new skills, modelling of successful others or scheduling of time and energy (Freund & Baltes, 2002). Depending on the (work) context, some optimisation strategies are especially relevant. Optimised time investment, for example, is important in the field of professional expertise (Wiese et al., 2000).

Thirdly, individuals can *compensate* for lost or soon to be lost resources by using alternative means in order to maintain a given level of functioning (Lang et al., 2011). Typical alternative means are external aids, seeking the help of others (Freund & Baltes, 2002) or changes in the allocation of one's effort (Freund & Baltes, 2002; Wiese et al. 2000). When facing resource loss, individuals may increase their effort in central areas of their lives while decreasing effort in other areas. If resources get scarce, for example when the end of their (working) lives is imminent, individuals increasingly focus on and invest their resources in pleasant interactions and situations (socioemotional selectivity theory; e.g., Carstensen, Fung, & Charles, 2003).

In other words, selection, optimisation, and compensation complement each other and have been shown to be key factors in the mastery of lifespan demands (e.g., Wiese et al. 2000). Originally developed as a theory to explain how individuals may maintain high levels of functioning until old age, SOC has also been investigated in its function as a buffer against the demands of work that helps employees to remain healthy and to maintain their performance levels even when resources become scarce (Abraham & Hansson, 1995; Bajor & Baltes, 2003; Schmitt, Zacher, & Frese, 2012; Yeung & Fung, 2009; Zacher & Frese, 2011), as well as in its function as a buffer against problems in balancing work and family or partnership (Baltes & Heydens-Gahir, 2003; Wiese et al., 2000; Young, Baltes, & Pratt, 2007).

3.2.1. Selection, optimisation, and compensation in the work context

Abraham and Hansson (1995) concluded from their study with 224 employees from different professions that SOC strategies,

especially elective selection and compensation, helped employees to maintain satisfactory levels of functioning. Yeung and Fung (2009) reported that compensation processes were related to higher job performance. Wiese et al. (2000) showed in a study with 206 young German employees from different professions that job satisfaction, emotional balance and the subjective feeling of success were influenced by SOC strategies.

Hence, there is some indication that SOC in the work context contributes to higher levels of well-being and job performance. Bajor and Baltes (2003) even show that SOC is a unique predictor of job performance. Schmitt et al. (2012) found that daily use of SOC reduced the effect of high problem solving demands on fatigue of employees. Effective self-regulation of resources by selection, optimisation, and compensation may, therefore, be beneficial to the well-being of teachers, too. We are thus going to investigate how teachers can make use of SOC strategies in order to allocate their time effectively which may help them to avoid emotional exhaustion and to remain healthy over the course of their careers.

3.2.2. The use of selection, optimisation, and compensation over teachers' career cycles

There is some indication for two different trajectories of age-related change in the use of SOC-related behaviour. Firstly, it is assumed that the older individuals are, the more they engage in SOC strategies (e.g., Wiese et al., 2000). Secondly, SOC may be especially relevant in certain phases of the (working) life (Freund & Baltes, 2002; Young et al., 2007). A model describing career stages of teachers (Huberman, Gronauer, & Marti, 1993) also illustrates stage-specific demands of teachers and provides indication for career stage-specific SOC use. In the following sections, we are going to provide evidence for both, linear and career-specific SOC use.

According to SOC research, higher age is accompanied by an imbalance of losses *versus* gains in resources (Freund, 2008). To adapt to this, individuals are assumed to constantly refine their knowledge and use of general SOC-related behaviour (Freund & Baltes, 2002) as well as the use of work-related SOC (Young et al., 2007). Thus, SOC use may improve steadily with increasing age. Wiese et al. (2000) also show a positive correlation between age and *selection* processes in the work context. With increasing tenure and the accompanying constant exposure to high work demands, teachers may make use of selection strategies by increasingly focussing on their core task of teaching and reducing some of the additional tasks such as administrative work or taking part in school projects. We therefore assume that older, experienced teachers make use of selection by focussing on teaching as the core aspect of their profession and give up on additional tasks.

Optimised time investment has been shown to be important, particularly in the field of professional expertise (Wiese et al., 2000) and thus also for teachers. Over the career cycle, teachers develop a rich knowledge base (Calderhead, 1996) and build up expertise. As one might expect, teachers can increasingly save time on tasks they regularly perform (e.g., individual lesson planning, correcting tests and homework). Bauer et al. (2007) compared teachers of different age groups with regard to the amount of time they spent on different aspects of the profession, and found some indication for an increasing optimisation process. Although younger and older teachers did not significantly differ in how they spent their working time in addition to giving or preparing lessons, a downward trend was noted for the latter. However, the results of that study might be biased due to an underrepresentation of younger teachers ($n = 27$ younger than 35 years; $n = 153$ who were 55 years or older). It may be argued that a larger database with data from a higher percentage of younger teachers would be necessary to investigate to what extent teachers are able to optimise their preparation of class or correction of tests. Yet, the crucial aspect in order to describe an *optimisation* process is which other tasks teachers invest their conserved time resources in. As COR theory (Hobfoll, 2002) states, an essential requirement for maintaining health even after extensive resource investment is to invest gained time resources in positive or rewarding tasks in order to gain new resources. Thus, we assume that experienced older teachers optimise the time spent on preparation of lessons and correction of tests and invest the time resources gained in restorative tasks instead.

Certainly, in order to *compensate* for lost resources, teachers could reduce their career ambitions (Wiese et al., 2000). While teachers with high career identification have been shown to invest more in tasks associated with their own qualification (Christ, Van Dick, Wagner, & Stellmacher, 2003), the teaching profession still tends to provide limited career opportunities. Reducing career ambitions over the career cycle might thus be beneficial to teachers. Another way of compensating would be for teachers to change the allocation of their overall effort (Wiese et al., 2000). Additionally, COR theory (Hobfoll, 2001) states that reallocating resource investment can protect individuals against further resource loss, and may help them to recover from loss or even gain new resources. Therefore, intensifying their efforts to manage the demands of their profession would be a beneficial compensation mechanism, especially for experienced teachers at the end of their careers who have already invested a great amount of time. We thus posit that older, experienced teachers compensate for resource investment by reducing their career ambitions and increasing their effort in order to manage the demands of teaching.

H1: The more experienced the teachers, the more SOC strategies they use in terms of

(H1a) focussing on teaching as the core aspect of their profession and giving up on additional tasks

(H1b) saving time on preparing lessons and correcting tests, and invest the time resources gained in restorative tasks instead (optimisation), and

(H1c) reducing their career ambitions and investing their effort (compensation) in order to manage the high demands of teaching.

On the other hand, some indication for career-specific use of SOC exists, especially highlighting the mid-career phase as a time of frequent SOC-related behaviour. Freund and Baltes (2002), for example, report that individuals in the middle of their career - a phase in which they are under particular pressure due to demands in the job as well as outside the job - are likely to use more SOC strategies. Young et al. (2007) also highlight the importance of SOC to individuals between the ages of 35 and 50 (indicating a curvilinear relationship). A study by Richter, Kunter, Klusmann, Lüdtke, and Baumert (2011), based on the same sample as this study, also provided evidence for non-linear effects by reporting quadratic trends for the use of training activities to target general skills.

To assess which career stages of teachers might be associated with increased SOC use, we draw on Huberman's model of teacher career stages (Huberman et al., 1993). Huberman postulates a set of five consecutive stages (survival and discovery, stabilisation, experimentation/activism and stocktaking, serenity and conservatism, and finally disengagement) which bring about different levels of demands. According to Huberman the first three years of teaching are often accompanied by a struggle to survive and manage the high demands of teaching and – once this has been successfully dealt with – by feelings of accomplishment and discovery. In this phase, SOC would be particularly useful in order to cope with the demands of teaching. After this teachers may

stabilise and become established in the profession. In the middle of their career, however, teachers may either become used to the demands of teaching and start to experiment with new materials and instructional strategies, or new demands can occur and teachers might reassess themselves and struggle with self-doubt. In the next phase, many teachers reach serenity and experience a greater sense of self-acceptance. Some of their colleagues at the same stage, however, become sceptical towards educational innovations and continue to struggle with the demands of teaching. Towards the end of their career, teachers tend to reduce their career ambitions and their commitment further, and prepare for retirement (Huberman et al., 1993). In order to accomplish this successfully, SOC strategies might also be beneficial. Huberman's model thus implicates frequent use of SOC at the beginning as well as towards the end of teachers' careers and infrequent SOC use in the middle career phases (curvilinear relationship).

Hence, we expect to also find curvilinear trends in the use of SOC over teachers' career cycles with increased SOC use at the beginning and towards the end of teachers' careers. Both linear and curvilinear effects in the use of SOC over teachers' career cycles will be tested in consecutive analyses.

H2: SOC use is especially high during the beginning as well as towards the end of teachers' careers.

4. Methodology

4.1. Participants

The current analysis is based on the first wave of the COACTIV study ("Professional Competence of Teachers, Cognitively Activating Instruction, and the Development of Students' Mathematical Literacy"; Kunter et al., 2007). This study was a part of the German extension to the 2003 cycle of OECD's Programme for International Student Assessment (PISA).[2] The sample consisted of 1939 in-service teachers of mathematics, science, and a range of other subjects (e.g., geography, or physical education; teachers in Germany are licensed for at least two subjects. Hence, a full list of all possible subject combinations cannot be provided). Participants were drawn from a nationally representative (in terms of state and school track) sample of 198 German secondary schools. Schools belonged to the academic track (so called *Gymnasium*) or the non-academic track (so called *Realschule* and *Hauptschule*). The school principal was the contact person at each school and was asked to administer the questionnaire to the teachers of the respective school. Participation in this study was voluntary and the teachers remained anonymous throughout the study.

The age of the teachers ranged from 25 to 65 years ($M = 47.4$, $SD = 9.4$) with a high percentage of older teachers (45.5% $\geq$ 50 years); teaching experience ranged from 1 to 44 years ($M = 20.7$, $SD = 10.6$). Half of the teachers (51.3%) were female and the majority of teachers (69.6%) worked full-time.

4.2. Measures

Emotional exhaustion as indicator of well-being was assessed by 5 items (e.g., "I often feel exhausted at school.") from the German translation of the Maslach Burnout Inventory (Enzmann & Kleiber, 1989). Participants were asked to rate their agreement on the emotional exhaustion items on a 4-point response scale (1 = *strongly disagree* to 4 = *strongly agree*). The internal consistency of the scale was good (Cronbach's Alpha = .80).

A list of teaching-related tasks (e.g., individual lesson planning, correction of tests and homework, meetings with students or parents, individual training, administrative tasks such as paperwork, documentation; for a full list see Table 2) was provided and teachers were asked how many hours they spent on each task in addition to their teaching hours within an average week. This list of tasks was specially designed for the COACTIV study. The number of tasks teachers engaged in was calculated from the data. On this basis selection processes were assessed. In order to evaluate optimisation processes, the time spent on each task was transferred into a ratio expressing the percentage of their time teachers spent on this task. We calculated ratios in order to examine differences in the relative proportions of time use instead of the absolute amounts of time spent on each of the single tasks.

Career ambitions (e.g., "I have high aspirations for my future career.") as well as readiness for investment of effort (e.g., "I spare no effort at work.") as indicators for compensation processes were both assessed with a well-established German scale, the Occupational Stress and Coping Inventory (AVEM; Schaarschmidt & Fischer, 1996). Participants were prompted by the instruction "We would like you to describe some of your typical behaviours, attitudes, and habits with respect to your working life," and were then asked to rate their agreement on both scales (4 items each) on a 5-point response scale (1 = *strongly disagree* to 5 = *strongly agree*). The internal consistencies of the career ambition (Cronbach's Alpha = .81) as well as the readiness for investment scale (Cronbach's Alpha = .80) were good.

Age as well as work experience were measured in years. Both were highly correlated ($r = .90$, $p < .05$), indicating that age and years of experience are almost interchangeable. The study by Young et al. (2007) examining work-related SOC behaviours also showed that age-related effects were more predominant. We conducted separate analyses with age and teaching experience as predictors, but the findings were equivalent and we therefore use the terms age and experience interchangeably.

Gender was controlled for in all analyses because women tend to use more SOC-related behaviour (Wiese et al., 2000). In our analyses gender was also significantly correlated with the number of hours spent on tasks as well as the ratios of time spent on individual lesson planning, correction of homework, meetings with students, and administrative tasks. We also controlled for the number of reduction hours of a teacher. Reduction hours are assigned if teachers take over tasks in the organisation of the school and thus have a higher rank in the school hierarchy. Controlling for the number of reduction hours makes sure that effects reported are not confounded by the level in the school hierarchy, which might be higher for older teachers.

4.3. Statistical analyses

To confirm previous results on working time, a set of different regression analyses was conducted with emotional exhaustion as dependent variable, controlling for gender and number of reduction hours. In a first regression analysis we tested whether the number of tasks and the total amount of time spent on them are relevant to teacher emotional exhaustion; in another analysis tasks which are more closely related to teaching were assessed. A third analysis tested the impact of additional tasks, and in a fourth analysis the compensation strategies were tested regarding their impact on emotional exhaustion.

In order to answer hypotheses H1a–c, single regression analyses for each ratio of time spent on a task, career ambitions, and

[2] The COACTIV project was funded by the German Research Foundation (DFG; BA 1461/2-2) as part of its Priority Program on School Quality (BIQUA).

readiness to invest effort with age as an independent variable were conducted. All analyses on age-related differences were also controlled for the number of reduction hours as proxy for engagement in school management, as well as gender.

To address hypothesis H2 polynomial regression analyses were conducted. In a polynomial regression analysis, which is a special case of multiple regression analysis (Kutner, Nachtsheim, & Neter, 2004); power functions of the predictors (x, x^2, etc.) are introduced in the regression to estimate curvilinear relationships between predictor and dependent variables. The shape of the function can be determined by testing the significance of each predictor in the model (x *versus* x^2). In the present analyses, we first introduced a linear term and then a quadratic term of age.

All analyses were calculated in Mplus (version 6; Muthén & Muthén, 1998–2010) with school as a cluster variable in order to account for the hierarchical structure of the data (teachers are nested in schools). The level of significance of all analyses was specified as $\alpha = .05$.

Missing data: The percentage of missing data ranged from 2.4% to 44.9%, with an average of 11% of missing data. Multiple imputation is increasingly accepted in the methodological literature to be superior to pairwise or listwise deletion. We therefore performed multiple imputation using the NORM software (version 2.03; Schafer, 2000). All data available on the study variables were included in the estimation of the missing values as auxiliary variables. We generated 10 imputed datasets which were then simultaneously analysed.

5. Results

Descriptive results show that teachers spend almost 32 h per week ($M = 31.87$, $SD = 13.61$) on an average of 11 teaching-related tasks ($M = 11.08$, $SD = 1.80$). As shown in Table 2, individual lesson planning is the most time-consuming task ($M = 9.52$, $SD = 4.50$) and accounts for almost one third (31.16%) of the time teachers spend on all teaching-related tasks. The second most time-consuming task is correction of tests ($M = 5.10$, $SD = 3.35$; 16.08%) followed by time spent on correction of homework ($M = 2.81$, $SD = 2.37$; 8.71%) and on administrative tasks ($M = 2.28$, $SD = 3.96$; 6.52%).

The teachers reported a moderate level of emotional exhaustion ($M = 2.12$, $SD = .64$) which increased slightly with age ($r = .07$,

Table 2
List of tasks, hours and relative time spent on each task in an average week.

	Hours per week *M* (*SD*)	Ratio in % *M* (*SD*)
Amount of time per week	31.87 (13.61)	–
Individual lesson planning	9.52 (4.50)	31.16 (12.29)
Preparation and correction of tests	5.10 (3.35)	16.08 (8.31)
Correction of homework and other tests	2.81 (2.37)	8.71 (5.57)
Documentation of students' performance	1.57 (1.49)	4.91 (2.84)
Meetings with students	1.27 (1.34)	3.84 (3.01)
Organisation of school projects	.91 (1.36)	2.81 (3.85)
Supervision of students	.82 (1.01)	2.53 (2.45)
Organisation of and participation in excursion	1.69 (1.96)	5.20 (4.37)
Meetings with parents	1.28 (1.38)	3.92 (2.52)
Individual training, reading of specialist literature	2.14 (2.11)	6.56 (4.59)
Administrative tasks	2.28 (3.96)	6.52 (9.73)
Attending school conferences	1.40 (1.45)	4.44 (2.60)
Other tasks	1.08 (1.26)	3.32 (2.77)
	M (SD)	
No. tasks	11.08 (1.80)	–

Table 3
Intercorrelations of variables; $N = 1939$.

	Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
1	Em. Exhaustion[1]	–																				
2	No. tasks	−.01	–																			
3	Total time[2]	.02	.32*	–																		
4	Lesson plan.[3]	.08*	−.37*	−.24*	–																	
5	Corr. tests[3]	.05	−.20*	−.02	−.10*	–																
6	Corr. homework[3]	.10*	−.01	.05*	−.20*	.07*	–															
7	Document. perf.[3]	.04	.05	.01	−.18*	−.09*	.05*	–														
8	Meet. students[3]	−.06*	.34*	.10*	−.33*	−.17*	−.01	.14*	–													
9	School projects[3]	−.06*	.31*	.03	−.19*	−.16*	−.10*	−.03	.08*	–												
10	Superv. students[3]	−.02	.43*	.04	−.22*	−.15*	−.02	.09*	.12*	.12*	–											
11	Excursions[3]	.01	.17*	.05	−.15*	−.13*	−.09*	.09*	−.01	.01	.12*	–										
12	Meet. parents[3]	.01	.23*	.08	−.33*	−.21*	−.03	.14*	.26*	−.04	.12*	.12*	–									
13	Ind. training[3]	−.03	−.07*	.08*	−.20*	−.15*	−.05	.00	−.01	.01	−.09*	−.09*	−.01	–								
14	Admin. tasks[3]	−.14*	.08*	.15*	−.40*	−.29*	−.25*	−.18*	.04	−.01	−.05*	−.18*	.06	.00	–							
15	Conferences[3]	−.01	.08*	−.03	−.25*	−.19*	−.06*	.12*	.11*	−.02	.07*	.10*	.21*	−.02	.02	–						
16	Ready. invest.[4]	.16*	.14*	.23*	.02	−.01	−.07*	−.08*	.01	.06*	−.05	−.09*	−.05	−.03	.12*	−.10*	–					
17	Career amb.[4]	−.11*	.13*	.17*	−.05	−.12*	−.08*	−.06*	.04	.10*	.02	−.06*	.04	.02	.19*	−.06*	.38*	–				
18	Age[5]	.07*	.04	.05*	−.28*	.07*	.05*	.01	.08*	−.03	.02	−.05	.08*	.05	.19*	.10*	−.02	−.24*	–			
19	Age square	.07*	.02	.05*	−.27*	.07*	.05	.01	.07*	−.03	.02	−.05	.07*	.05	.19*	.10*	−.03	−.27*	–	–		
20	No. red. hours[2]	−.14*	.06*	.20*	−.34*	−.20*	−.16*	−.10*	.14*	.02	.01	−.14*	.17*	.09*	.59*	.07*	.12*	.22*	.23*	.23*	–	
21	Gender[6]	.01	−.14*	−.03	−.07*	.04	−.10*	−.01	−.09*	.07	−.11*	−.12*	−.07*	.11*	.18*	.01	−.07*	−.01	.21*	.22*	.15*	–

Note. [1] scale from 1 = "*strongly disagree*" to 4 = "*strongly agree*", [2] in hours, [3] relative time spent on task in %, [4] scale from 1 = "*strongly disagree*" to 5 = "*strongly agree*", [5] in years, [6] 1 = *female*, 2 = *male*.
$^*p < .05$.

Table 4
Predicting the emotional exhaustion of teachers by relative time spent on different tasks (results of four different regression analyses; controlled for gender and number of reduction hours).

	B (*SE*)	ß
No. tasks	−.01 (.01)	−.02
Total amount of time per week[1]	.01 (.01)	.06*
R^2		.02
Individual lesson planning[3]	.01 (.01)	.07*
Preparation and correction of tests[3]	.00 (.01)	.03
Correction of homework and other tests[3]	.02 (.01)	.10*
R^2		.03
Documentation of students' performance[3]	−.01 (.01)	.02
Meetings with students[3]	−.02 (.01)	−.06*
Organisation of school projects[3]	−.01 (.01)	−.06*
Supervision of students[3]	.00 (.01)	−.01
Organisation of and participation in excursions[3]	−.01 (.01)	−.03
Meetings with parents[3]	.02 (.01)	.05
Individual training, reading of specialist literature[3]	−.01 (.01)	−.03
Administrative tasks[3]	−.01 (.01)	−.11*
Attending school conferences[3]	.00 (.01)	.00
R^2		.04
Readiness for investment[4]	.19 (.02)	.24*
Career ambitions[4]	−.14 (.02)	−.17*
R^2		.08

Note. *B* = unstandardised regression coefficient, *SE* = standard error of unstandardised regression coefficient, ß = standardised regression coefficient, R^2 = variance explained by the model, [1] in hours, [3] relative time spent on task in %, [4] scale from 1 = "*strongly disagree*" to 5 = "*strongly agree*".
*$p < .05$.

Table 5
Predicting the relative time spent on each task by teachers' age (controlled for gender and number of reduction hours).

	Model 1			Model 2		
	B (*SE*)	ß	R^2	*B* (*SE*)	ß	R^2
No. tasks						
Age	.01 (.03)	.06	.03	.00 (.01)	.01	
Age^2				.00 (.00)	−.12*	.04
Total amount of time per *week*[1]						
Age	.02 (.03)	.01	.05	.02 (.04)	.01	
Age^2				.00 (.00)	.00	.05
Individual lesson planning[3]						
Age	−.28 (.03)	−.22*	.16	−.24 (.04)	−.19*	
Age^2				.01 (.00)	.08*	.16
Correction of homework and other tests[3]						
Age	.06 (.02)	.11*	.04	.07 (.02)	.11*	
Age^2				.00 (.00)	.00	.04
Documentation of students' performance[3]						
Age	.01 (.01)	.03	.01	.01 (.01)	.04	
Age^2				.00 (.00)	.02	.01
Meetings with students[3]						
Age	.02 (.01)	.07*	.04	.01 (.01)	.04	
Age^2				.00 (.00)	−.08*	.04
Organisation of school projects[3]						
Age	−.02 (.01)	−.05	.01	−.02 (.01)	−.06	
Age^2				.00 (.00)	−.03	.01
Administrative tasks[3]						
Age	.05 (.02)	.04*	.36	.04 (.03)	.04	
Age^2				.00 (.00)	.00	.36
Readiness for investment[4]						
Age	−.00 (.00)	−.03	.02	−.01 (.00)	−.06*	
Age^2				.00 (.00)	−.07*	.03
Career ambitions[4]						
Age	−.03 (.00)	−.31*	.14	−.03 (.00)	−.31*	
Age^2				.00 (.00)	.00	.14

Note. *B* = unstandardised regression coefficient, *SE* = standard error of unstandardised regression coefficient, ß = standardised regression coefficient, R^2 = variance explained by the model, [1] in hours, [3] relative time spent on task in %, [4] scale from 1 = "*strongly disagree*" to 5 = "*strongly agree*".
*$p < .05$.

$p < .05$, see Table 3). In a first confirmatory step of our analyses, we tested if the total time spent on all tasks, the total number of tasks as well as the ratios of time spent on each single task were indeed relevant to the teachers' emotional exhaustion.

As one might expect, the more time teachers spent on all tasks the more exhausted they were (ß = .06, $p < .05$; see Table 4). Interestingly, the total number of tasks did not predict the level of emotional exhaustion. In a multiple regression analysis that followed, we tested whether the ratios of time spent on those tasks which are more closely related to teaching were relevant to teachers' emotional exhaustion. The results show that the ratios of time spent on individual lesson planning (ß = .07, $p < .05$) and correction of homework (ß = .10, $p < .05$) are also positively associated with teachers' emotional exhaustion while the relative time spent on preparation and correction of tests is not. In a further analysis we tested whether the ratios of time spent on additional tasks were also relevant to teachers' emotional exhaustion. Interestingly, results show that the relative time spent on administrative tasks (ß = −.11, $p < .05$), the organisation of school projects (ß = −.06, $p < .05$) as well as meetings with students (ß = −.06, $p < .05$) were negatively associated with emotional exhaustion. The more of their time teachers spent on these tasks the less exhausted they were.

The bivariate correlations listed in Table 3 provide a first indication of differences in teachers' SOC according to age. To address hypotheses H1a to H1c, we calculated single regression analyses with age as independent variable, and in order to address hypothesis H2 we calculated polynomial regression analyses by entering age square in each analysis.

Results show that no linear age-related effects in the overall number of tasks (ß = .06, $p > .05$) nor in the average time spent on all tasks (ß = .01, $p > .05$) occur. Hypothesis H1a thus has to be rejected. A significant quadratic trend was only obtained for the total number of tasks ($ß^2_{age} = -.12$, $p < .05$; see model 2 in Table 5). In other words, teachers in different phases of their careers do not differ in how much time they spend on all tasks per week. Yet, teachers at the beginning and the end of their careers engage in fewer tasks than their colleagues in the middle phase (result supports hypothesis H2; see Fig. 1).

Next, we addressed hypothesis H1b and assessed whether teachers have the opportunity to optimise those teaching-related tasks which take up most of their time over the career cycle: the ratios of time spent on preparing lessons as well as correcting tests and homework. A linear effect was obtained only for the ratio of time spent on correction of homework (ß = .11, $p < .05$). This shows that experienced teachers invest even more of their time in correction of homework and no cumulative gain in time seems to take place (hypothesis H1b has to be rejected). Yet, we found a quadratic trend in the relative time spent on lesson planning ($ß^2_{age} = .08$, $p < .05$; see model 2 in Table 5). In other words, whereas teachers in the middle phase of their careers spend less of their time on lesson planning than beginning teachers, their colleagues at the end of their careers are not able to reduce this ratio further (supporting hypothesis H2).

In order to assess optimisation it is crucial to investigate which aspect of their profession teachers spend their conserved time on instead. Interestingly, we found a linear trend for administrative tasks (ß = .04, $p < .05$), which means that experienced teachers spend more of their time on administrative tasks even after controlling for number of reduction hours as a proxy for duties in school management (see Fig. 2). In addition, we found a quadratic

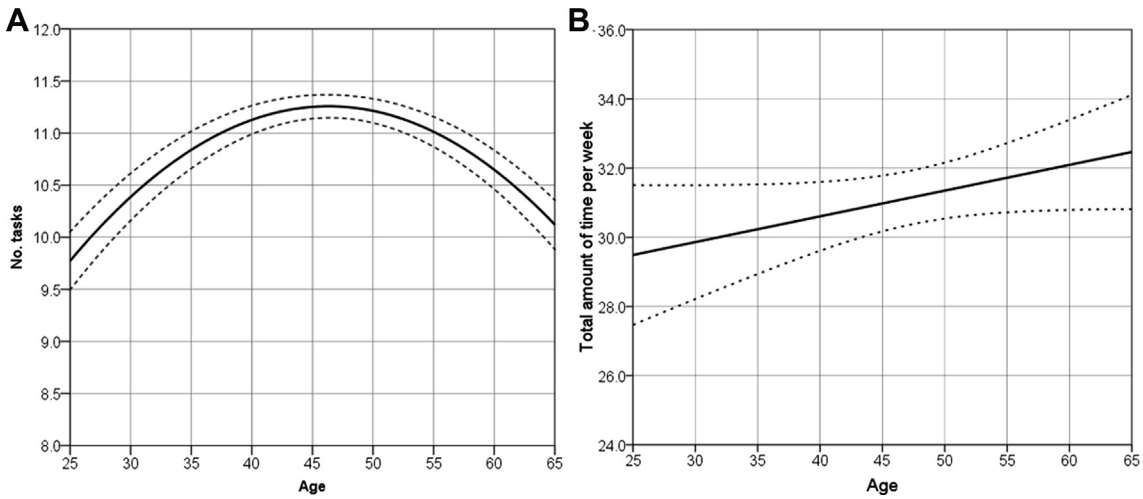

Fig. 1. Number of tasks (A) and total amount of time per week spent on all tasks (B) over the career cycle. *Note*. The solid line indicates the function predicted by the regression analysis, the dashed lines delimit the 95% confidence interval.

trend for meetings with students ($ß^2_{age} = -.08$, $p < .05$), suggesting that especially the teachers at the beginning but also those at the end of their careers spent less of their time in meetings with students than their colleagues in the middle phase of their careers. Thus hypothesis H2 is supported.

We then examined age-related effects in the readiness for investment of effort as well as teachers' career ambitions. The teachers reported a moderate level of readiness to invest effort ($M = 2.96$, $SD = .80$) or career ambitions ($M = 2.78$, $SD = .79$). Results show that career ambitions and age form a linear

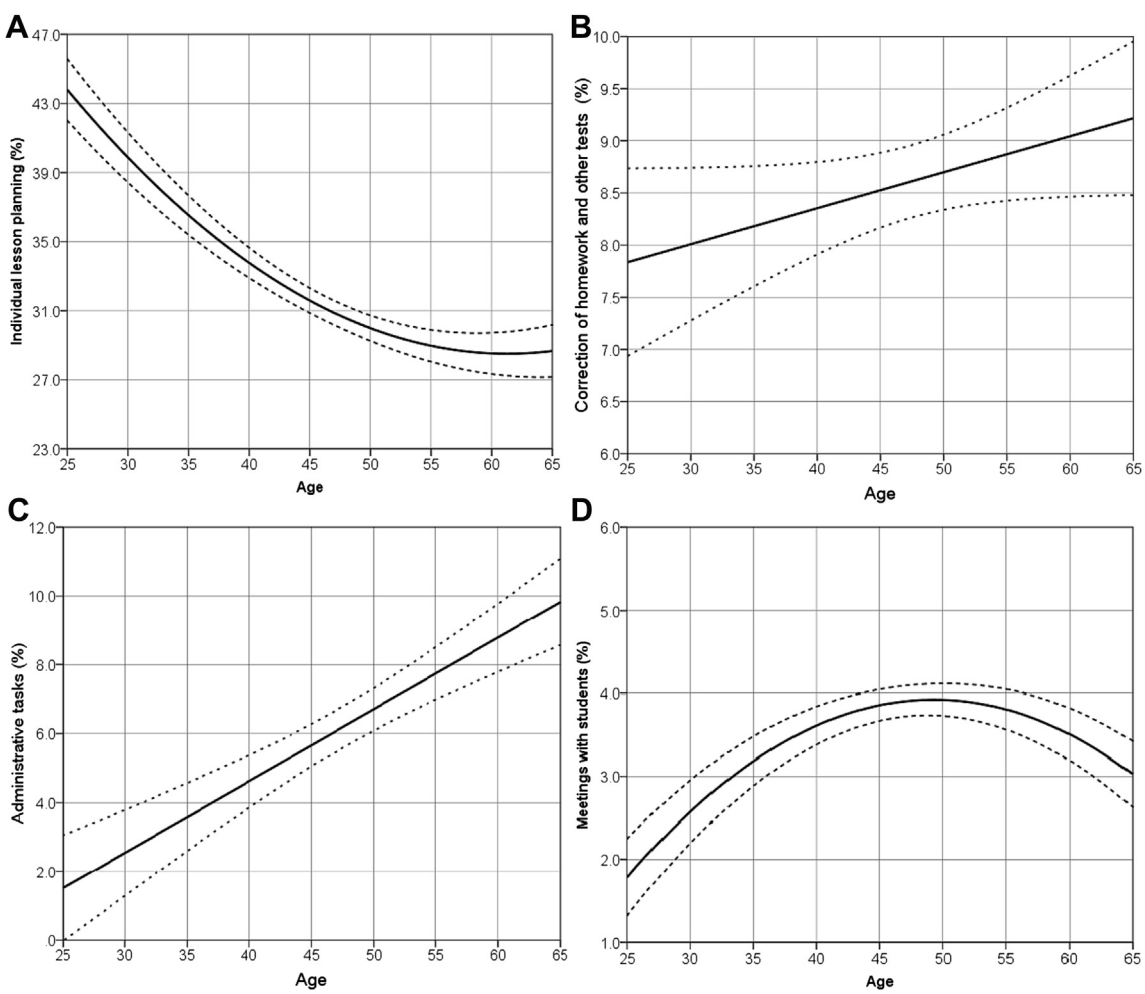

Fig. 2. Ratios of time spent on individual lesson planning (A), correction of homework and, other tests (B), administrative tasks (C) and on meetings with students (D). *Note*. The solid line indicates the function predicted by the regression analysis, the dashed lines delimit the 95% confidence interval.

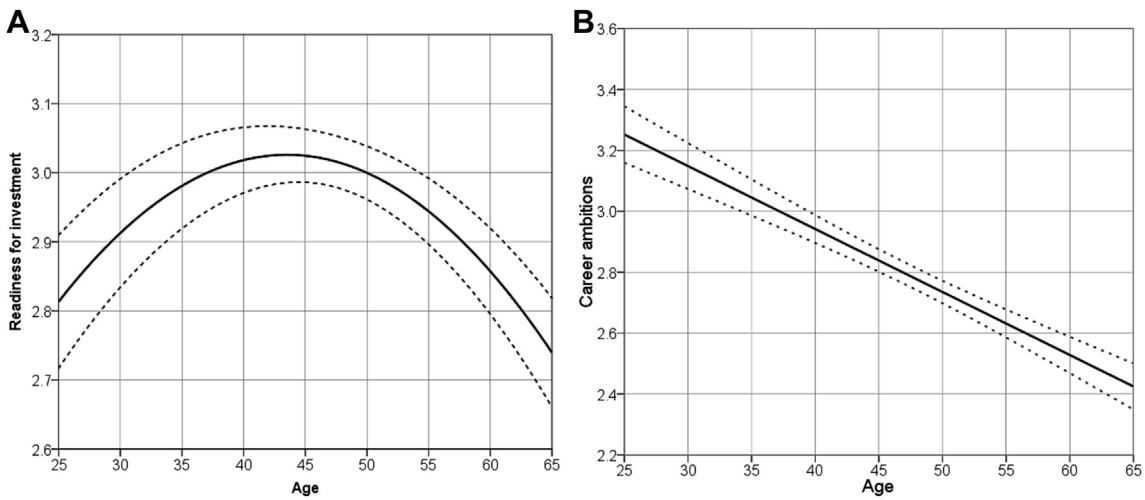

Fig. 3. Readiness for investment (A) and career ambitions (B) over the career cycle. *Note*. The solid line indicates the function predicted by the regression analysis, the dashed lines delimit the 95% confidence interval.

relationship ($\beta = -.31$, $p < .05$, see Fig. 3), suggesting that older, experienced teachers invest fewer resources in their careers (supporting hypothesis H1c). A significant quadratic trend was found for their readiness to invest effort ($\beta^2_{age} = -.07$, $p < .05$), which suggests that beginning teachers as well as teachers at the end of their careers invest less effort than their colleagues in the middle career phase (supporting hypothesis H2).

6. Discussion

In the current discussion on teacher working time one of the main concerns is how teacher working time and the subjective workload they experience can be reduced in order to keep effective teachers in the profession (OECD, 2005). Research has shown that perceived workload is indeed relevant for teacher stress and burnout (Rudow, 1999). Less evidence for the association of teachers' working time (as objective demand) and stress and burnout exist. Yet, results on other employees have demonstrated that in addition to perceived workload working time can be considered a demand which is associated with reduced well-being (Skinner & Pocock, 2008). So far, studies have shown that teachers from Western countries (Bruno et al., 2012; Butt & Lance, 2005; Ingvarson et al., 2005; Lacroix et al., 2005; OME, 2005) indeed work long hours.

In our study we not only quantified the number of working hours of secondary teachers from a representative school sample in Germany, the number of tasks engaged in and which tasks are especially demanding, but also assessed how teachers divide their time among the broad number of teaching-related tasks as indicator of teachers' self-regulation capacity (third level according to Ingvarson et al., 2005). Our results are based on a German sample, but they can be transferred to other countries in which teachers work comparably long hours. For a theoretical framework of the assessment of teachers' self-regulation capacity, we draw on conservation of resources (COR) theory (Hobfoll, 2002) as well as on the selection, optimisation, and compensation (SOC) model (Baltes & Baltes, 1990; Lang et al., 2011).

First, results of a descriptive analysis of teachers working time show that the 1939 secondary school teachers in our sample work almost 32 hours per week in addition to teaching hours and engage in an average of 11 tasks. The amount of time the teachers invest in all the tasks associated with teaching may seem very high. However, bearing in mind that teachers in Germany work up to 57.3 hours in an average week (Lacroix et al., 2005), US teachers work approx. 54 hours (Bruno et al., 2012) and UK teachers work up to 49.9 hours per week (Butt & Lance, 2005), this number seems to be plausible. It also indicates that the teachers in this sample are indeed under severe pressure as far as time resources are concerned. Of course, the most time-consuming tasks are individual lesson planning, correction of tests as well as correction of homework, which are closely related to teaching. Additional tasks still account for 44% of teachers' working time. Of these, administrative tasks which account for 6% of overall time are more time-consuming than meetings with students or the organisation of school projects. These figures correspond to the results from UK secondary school teachers reported by Gunter et al. (2005) as well as to the results from US teachers (Bruno et al., 2012).

Next, we correlated teachers' working time (in addition to teaching), number of tasks engaged in (as indicator of job complexity) as well as each single task in particular to their emotional exhaustion in order to provide evidence on how straining each might be. In line with Skinner and Pocock (2008) we also found that the total amount of time teachers spend on all tasks is indeed relevant to their emotional exhaustion. At first, this result may not be surprising. Yet, it provides new insights as instruments less prone to common method variance were employed. Indeed, we only found a small effect ($\beta = .06$) which is, however, similar to the effects shown in the study by Skinner and Pocock (2008).

Contrary to results by Smith and Bourke (1992), results of the present study on the link between single tasks and emotional exhaustion show that the relative time spent on the most time-consuming tasks (individual lesson planning and correction of homework) was not found to be associated with higher emotional exhaustion. As one might expect, the ratios of time spent on other less time-consuming tasks such as meetings with students or school projects are associated with less emotional exhaustion. Apart from this, not all tasks seem to be as demanding as might be expected from the literature (Chaplain, 2008; Rudow, 1999). Overall it can be said that the more of their time teachers spend on administrative tasks (e.g., paperwork, documentation) the less exhausted they are.

Second, a core aspect of our study was to investigate whether teachers invested their time resources differently according to their age or career phase. Based on SOC theory (Baltes & Baltes, 1990; Lang et al., 2011), which states that with increasing age individuals refine their resource allocation by using selection, optimisation, and compensation behaviours (Freund & Baltes, 2002), we assumed that over the career cycle teachers were also making use of such

strategies. We postulated that the older and more experienced teachers are the more they reduce the number of tasks they engage in (which would indicate a selection process), optimise their time investment by reducing the relative time spent on the most time-consuming tasks (such as preparation of lessons or correction of homework) and use the free time resources for less exhausting tasks. They may also compensate for resource losses by being prepared to invest more effort or by reducing their career ambitions. From the literature on SOC (Freund & Baltes, 2002; Young et al., 2007) we expected not only linear effects but also curvilinear trends highlighting the importance of certain career phases. We indeed found evidence for both, linear as well as curvilinear trends which highlight that especially the teachers at the end of their careers and to some extent also beginning teachers were making use of SOC-related behaviours.

The curvilinear trend found for the number of tasks teachers engage in seems to support the assumption that teachers at the very beginning and the very end of their career engage in fewer tasks and are evidently making use of a selection strategy. According to Huberman et al. (1993) the first years of teaching can be characterised as a survival phase associated with the struggle to become an effective teacher, and very young teachers might benefit from concentrating on fewer tasks. Meanwhile, by the very end of their career teachers tend to disengage from their profession (Huberman et al., 1993), which might be reflected in our results. We cannot distinguish between elective and loss-based selection but it can be assumed that younger teachers use elective selection while the effect in older teachers may be driven by a loss of resources (as suggested by Lang et al., 2011). Future studies will have to investigate this distinction.

Results indicate that an optimisation process in the sense of SOC theory takes place. Unsurprisingly, older and very experienced teachers were able to reduce the relative time spent on individual lesson planning which is according to our results associated with increased emotional exhaustion. What is interesting, however, is that they shifted their priorities towards administrative tasks or meetings with students, which are both negatively associated with emotional exhaustion. These results were controlled for the number of reduction hours teachers receive (as an indicator for school management responsibilities) and are thus not just a reflection of teachers moving up in the school hierarchy.

There may be two possible explanations for this optimisation process. First, individuals who perceive their time as limited increasingly focus on positive interactions or activities and actively avoid interactions or activities considered negative (socioemotional selectivity theory; e.g., Carstensen et al., 2003). This might also be the case for experienced teachers who prefer to invest time resources they saved on demanding aspects of their profession and engage in undemanding and potentially rewarding tasks such as meetings with students instead. Second, this process could also be interpreted as task differentiation: Being able to switch between tasks gives individuals the opportunity to recover from more demanding tasks and concentrate on others instead, which has a potentially positive effect on their well-being (Hackman & Oldham, 1976). This optimisation process, however, seems to lose importance towards the very end of teachers' careers, as the curvilinear trends for the relative time spent on individual lesson planning and meetings with students suggest. Again, if retirement is imminent (Huberman et al., 1993), disengagement might be the reason for this.

As far as compensatory processes are concerned, we postulated that teachers would maintain the effort invested in the management of the demands associated with teaching whilst reducing career ambitions. The older the teachers were, the less ambitious they were about their careers, which we interpret as a compensation process because it would be highly inefficient for teachers to still invest in a career when career opportunities are becoming more and more limited. However, results of the polynomial regression analysis show that beginning teachers and especially their colleagues at the end of their careers are most likely to decrease their effort and make no use of this potential compensation strategy. Richter et al. (2011) who investigated the uptake of learning opportunities by teachers of different ages based on the same sample found similar trends. In that study, older teachers made no more use of help seeking than of any other potential compensation process. The authors report that in fact older teachers cooperate less frequently than their younger colleagues.

Selection, optimisation, and compensation processes are intertwined, as Lang et al. (2011) highlight: "These three principles conjointly serve to enhance and secure the individual's potential for positive development outcome" (p. 63). However, our results indicate that the teachers in this sample were only able to make use of some aspects of SOC as potentially beneficial self-regulation strategies.

7. Practical and theoretical implications

In our study we provided some evidence on how teachers at different stages of their careers allocate their time resources to the different tasks associated with teaching. Teachers towards the end of their careers and, to a lesser extent, beginning teachers make use of SOC-related behaviour which allows them to allocate their time resources more effectively which in turn might act as a buffer against the high demands of teaching. Teaching hours are formally specified in Germany as well as in most other OECD countries (OECD, 2011). Thus, teachers have some degree of freedom as to how much time they allocate to tasks associated with teaching. The teachers in our study make use of this freedom to some degree. Yet, under the current circumstances, possibilities for selection, optimisation, and compensation are limited. As a practical implication of our study we conclude that professional development programmes for teachers should take into account the fact that teachers need more flexibility in allocating their time resources: by being able to prioritise some aspects of their profession (selection), by optimising resource allocation to tasks which are not emotionally exhausting, by being provided with opportunities for personal growth to promote the development of a so called gain cycle (COR theory; Hobfoll, 2002), and by compensating for resource loss.

The increased use of information and communication technology may also act as a compensatory aid to reduce the high workload of teachers (Selwood & Pilkington, 2005). Compensatory principles were already applied in the so called Pathfinder Project which aimed at reducing the workload of UK teachers (Butt & Lance, 2005). In this project, schools were provided with consultancy support, principals were trained in change management, additional teaching assistants were hired, ICT hardware and software was provided and funding or bursarial training of school managers was provided. This resulted in a decrease in the overall workload of those 311 teachers who took part at both measurement points in 2002 and 2003. We conclude that increased opportunities for all three behaviours, selection optimisation, and compensation, may help teachers to maintain their well-being and remain in the profession for longer, which is a major concern according to the OECD report (2005).

Our study also has some theoretical implications. Many studies measure SOC using different versions of the well-established SOC scale (e.g., Freund & Baltes, 2002) while some use different means: in a study by Freund and Baltes (2002) individuals preferred SOC-related over non–SOC–related proverbs when asked to describe ways of life management. Meanwhile Li, Lindenberger, Freund, and Baltes (2001) used a behavioural-observational measure in which individuals had to fulfil a dual task (walking and memorising

simultaneously). Overall, it seems fruitful to add to the research on SOC by applying measures other than the SOC questionnaire.

However, apart from these strengths the present study has some limitations. It might be possible that teachers, in retrospect, overestimated the time they spend on their tasks. In order to provide more reliable data, observational or diary studies would be necessary (see Dunn & Shriner, 1999). In their study on self-reported time use of special education teachers, Vannest and Parker (2010) that the greatest gains in accuracy of time use were achieved with data collected over two to five days, with lesser gains when collected over about 11 days. In self-report studies several measurement points would thus provide more reliable data.

It is possible that teachers who started work in the 1960s have different assumptions about their profession and would prioritise different tasks than teachers who have entered the profession only recently. Following the same cohort of teachers over their careers could rule out such effects and would also allow for describing patterns of development. For this purpose, longitudinal studies would be necessary. In our study with its cross-sectional nature we can only make statements on differences between teachers of different age and we were careful in providing indications for changes over the career cycle.

A positive school climate may also contribute to teachers' readiness to engage in additional activities. The social environment has been investigated with a view to its influence on extra-role behaviours of teachers: a study by Somech and Ron (2007) has shown that a school climate in which members of a school express solidarity with each other contributes to teachers' readiness to engage in extra-role activities.

Moreover, we cannot rule out the healthy worker effect which is often found in comparable studies (Frese & Semmer, 1986). According to this effect older workers with health problems are less likely to take part in studies and the number of healthy older workers is thus overrepresented. Such a process is indeed likely because of the significant correlation of exhaustion and age square which indicates that the linear increase in emotional exhaustion over the age groups levels off in the oldest teachers (>60 years).

Furthermore, the effects we report are relatively small compared to questionnaire studies in which teachers are asked about their demands and well-being. First, this argument suggests that the proportion of time actually spent on a certain task is less strong a predictor than the individual's perception of a task. The effects in other studies are often overestimated due to a common method bias which is less likely to have occurred in our study in which we employed other measures of SOC-related behaviour in assessing how much time teachers spent on different tasks. Finally, smaller effects were to be expected in such a large sample due to the large error variance in the data. The relatively small effects on emotional exhaustion could equally be explained by the fact that the nature of overtime work should also be taken into account. If individuals have control over overtime work and do it voluntarily they are less likely to become fatigued (Beckers et al., 2008). Moderate overtime work may thus be less problematic. The authors reason that compulsory overtime may be partly offset by compensation processes which we found some indications for. Future studies should also not only rely on self-report data but also include other data sources such as co-workers or head teachers in order to reduce self-report bias.

One might argue that teachers may be able to use their holidays in order to recover from the demands of their profession. Results by Kühnel and Sonnentag (2011) indicate that teachers indeed perceive such an effect immediately after their holiday. Yet, the authors conclude that such a beneficial effect is likely to fade out quite quickly if teachers are confronted with the same demands as before their holiday.

To sum up, teachers are under pressure as far as their time resources are concerned and teachers tend to work long hours. However, only some teachers are severely burned out. Hence, general assumptions about the health and well-being of teachers should be made carefully. Our study showed that SOC strategies might help teachers to remain healthy until regular retirement age. Over the career of teachers some potential for SOC-related behaviour exists. Younger and older teachers engage in fewer tasks (selection), older teachers save time on more demanding aspects of their profession and invest their time in less demanding tasks instead (optimisation), and older teachers have fewer career ambitions (compensation). However, teachers can only make use of these strategies to a limited extent. This leads us to recommend that teachers of different age groups should be allowed to develop their own special profile with varying proportions of time allocated to different tasks, according to their resources.

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